2020秋高数上期中试题(回忆版)

1. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) Let $f(x)=|x|\sin x$. The greatest value of n, for which $f^{(n)}(0)$ exist, is _____.

(A) 0 (B) 1 (C) 2 (D) 3

(2) If $\lim_{x\to\infty}\left(\frac{x^2+1}{x+1}-ax-b\right)=\frac{1}{2}$, then the values of a,b are _____.

(A) $a=1, b=-\frac{3}{2}$ (B) $a=-1, b=\frac{3}{2}$ (C) $a=-1, b=1$ (D) $a=1, b=-1$.

(3) Which one of the following functions is not differentiable at $x=0$?

(A) $f(x)=|x|\sin|x|$ (B) $f(x)=|x|\sin\sqrt{|x|}$ (C) $f(x)=\cos|x|$ (D) $f(x)=\cos\sqrt{|x|}$

(4) The average value of function $g(x)=x^2+6$, for $0\le x\le 6$ is _____.

(A) 12 (B) 18 (C) 16 (D) 10

(5) What is the derivative of $f(x)=\frac{1-\sin x}{1+\sin x}$ at $x=\frac{\pi}{6}$?

(A) $\frac{4\sqrt{3}}{9}$ (B) $-\frac{\sqrt{3}}{3}$ (C) $-\frac{4}{3\sqrt{3}}$ (D) $\frac{1}{3}$

2. Please fill in the blank for the questions below.

(1) The integration $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^5x\cos^3x\,dx$ equals _____.

(2) If f is continuous and $\int_0^{x^3-1}f(t)\,dt=x$, then $f(7)=$ _____.

(3) If $f'(x)=(x+\frac{1}{x})^2$ and $f(1)=1$, then $f(x)=$ _____.

(4) A particle is moving on the sphere $x^2+y^2+z^2=13^2$. While $t=t_0$, $x(t_0)=3, y(t_0)=4, z(t_0)=12$, $y'(t_0)=3$, then $z'(t_0)=$ _____.

(5) $\lim_{s\to a}\frac{\sqrt{s^2+1}-\sqrt{a^2+1}}{s-a}=$ _____.

3. Find the limits (DO NOT apply l'Hôpital's Rule.)

(1) $\lim_{x\to\infty}\frac{\sqrt{3+x}-\sqrt{x+1}}{x^2+x-2}$

(2) $\lim_{x\to 0}\frac{\cos x-\sec^2x}{x\sin x}$

4. Evaluate the integral.

(1) $\int_0^{2\pi}|\sin^2x-\cos^2x|\,dx$

(2) $\int_0^1(x+2)\sqrt{1-x^2}\,dx$

5. Let $f(x)=\frac{x^3+x-2}{x-x^2}$.

(1) Indentify the inflection points and local maxima and minima of the function that may exist.

(2) Identify the horizontal, vertical, and oblique asymptotes that may exist.

(3) Graph the function.

6. (1) Find $\frac{dy}{dx}$ if $y(x)=\int_1^{1+2x}\sqrt{t^2-1}\,dt$, $x>0$.

(2) Find the equation of the line that is tangent to the curve $x^2-y^2=9$ at point $(5,-4)$.

7. Find the area of the region bounded by curves $y=x^2$ and $y=2x-x^2$.
8. Find the volume of the solid generated by revolving the region bounded by $y=2x-1$, $y=\sqrt{x}$ and y-axis about the line $x=-1$.
9. Assume that f is continuous on $[0,1]$. Show that there exists a number $c\in(0,1)$ such that $f(c)=\int_0^1 f(x)\,dx$.